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Department of Electronics & Communications
Engineering



Title:

**“Emergency Alert and GPS tracking device
for Women safety”**

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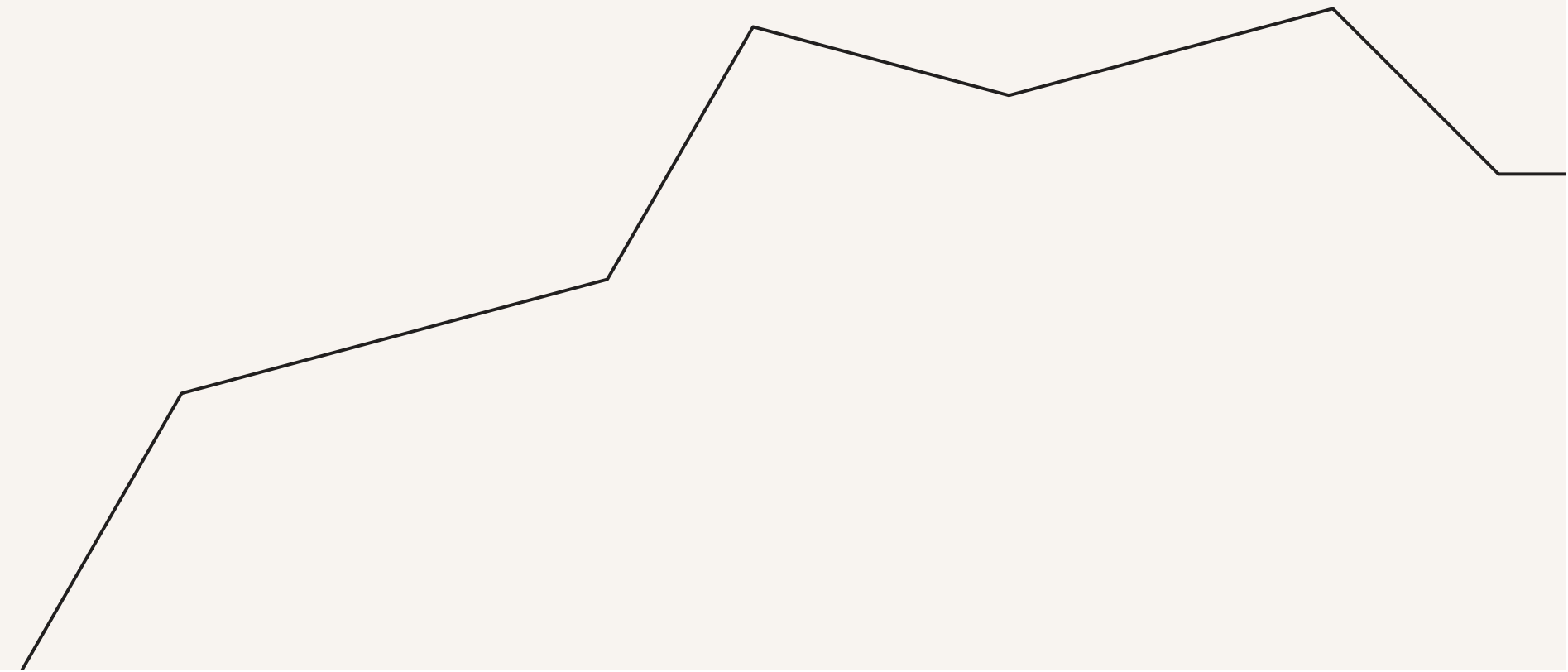
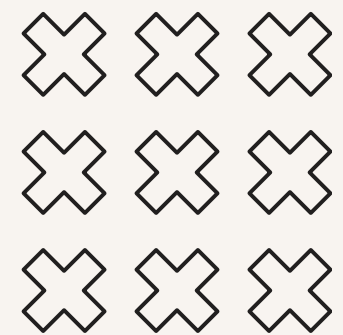
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Empowering Women: The Role of Emergency Alerts and GPS Tracking in Personal Safety

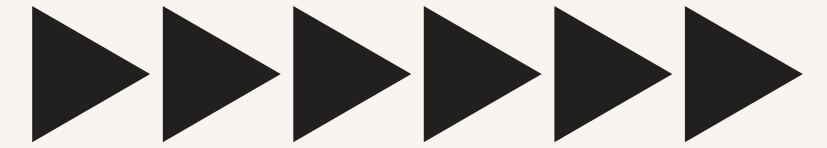
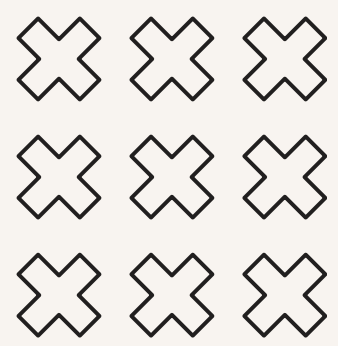




INTRODUCTION:

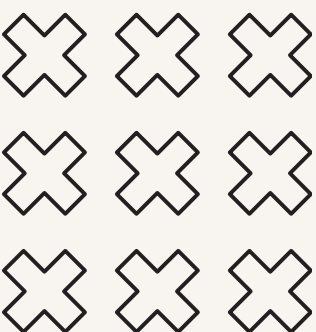
Increasing incidents of harassment and violence against women highlight the need for quick, effective safety solutions. Traditional methods (like phone calls or alerts) may not always be accessible in emergencies, creating a demand for reliable,





The Importance of Personal Safety

Personal safety is crucial for **women's empowerment**. Feeling safe allows women to pursue their goals and ambitions without fear. By utilizing **technology**, we can create safer environments and support systems that enhance women's confidence and independence.





Literature survey:

1. Introduction to Women's Safety Concerns
Increasing safety concerns for women globally, especially in urban areas.
Rising cases of harassment and violence call for innovative technological interventions.
2. Importance of Emergency Alert and GPS Tracking Systems
Enables quick response and real-time tracking during emergencies.
Provides a sense of security for women and a tool for their loved ones to ensure safety.
3. Existing Technologies in Women's Safety
Mobile-based SOS alert apps (e.g., bSafe, My Safetipin).



Wearable devices with panic buttons or alarms (e.g., smart rings, bracelets, keychains).
GPS-enabled tracking systems integrated with mobile apps.

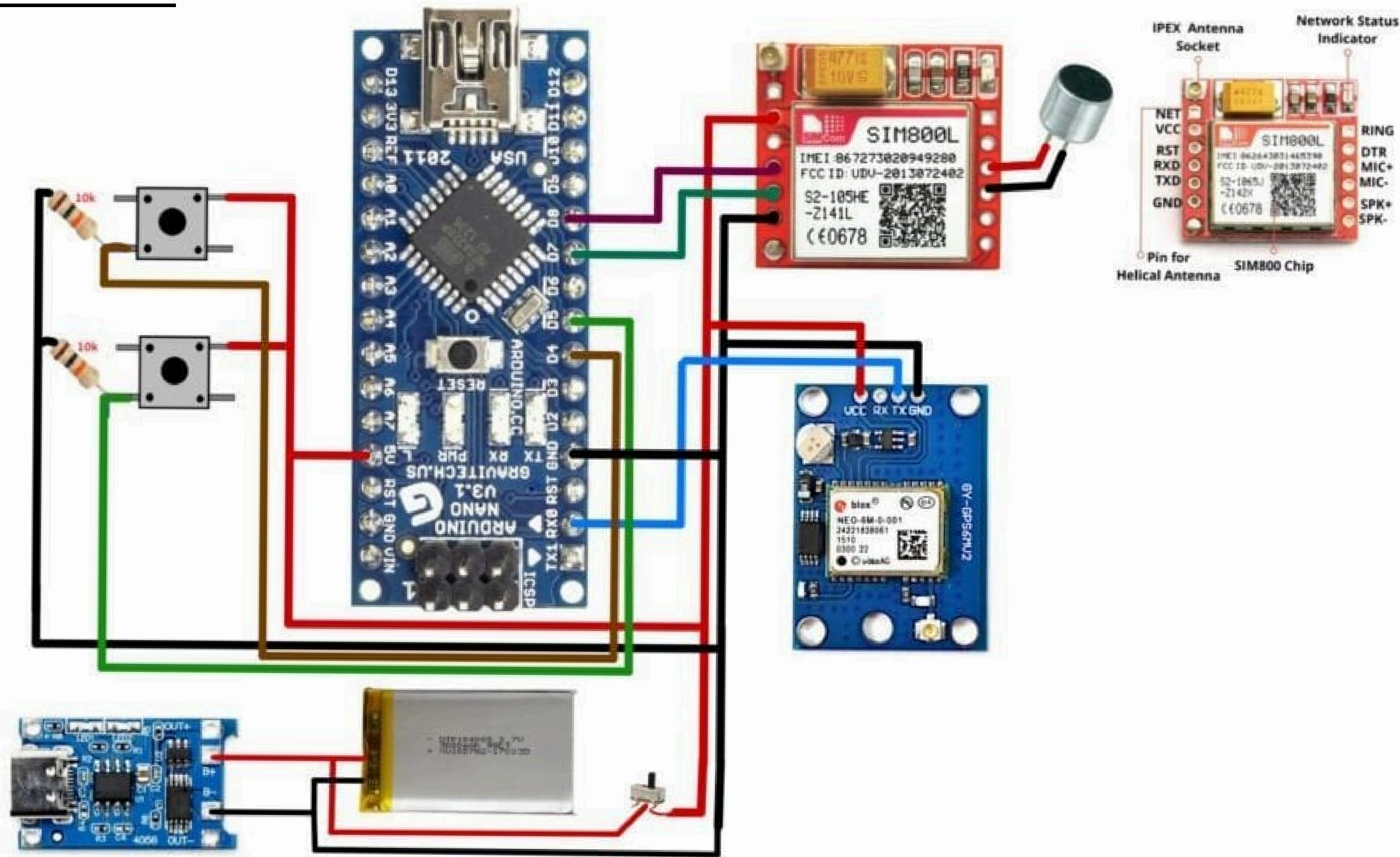
4. Features of Effective Women's Safety Devices

- ^ pepper spray**
- ^ personal alarms**
- ^ stun guns**

objectives:

- I. Enhance Personal Security: Provide women with a reliable tool to ensure their safety and peace of mind.
- II. Quick Emergency Alerts: Enable rapid notification to pre-selected contacts or authorities in case of an emergency.
- III. Real-Time GPS Tracking: Offer live location tracking to facilitate quick assistance and rescue efforts.
- IV. Easy-to-Use Interface: Ensure a user-friendly design that can be easily accessed and activated in stressful situations.
- V. Compact and Portable: Design a lightweight and discreet device that can be easily carried or worn.

Circuit Diagram:



Materials Used:

Arduino Nano

SIM800 Modem (Subscriber Identity Module)

NEO6M GPS module (Global Positioning System)

433 MHZ RF Transmitter and Receiver

Button

Battery

Breadboard

Jumpers

Excepted Result:

When the emergency button of the device is triggered once, the GSM sends the location of the victim to the police.



Benefits of GPS Tracking

Using **GPS tracking** can enhance personal safety by allowing friends or family to monitor a person's location. In emergency situations, this can facilitate faster responses and provide crucial information to authorities, potentially saving lives.

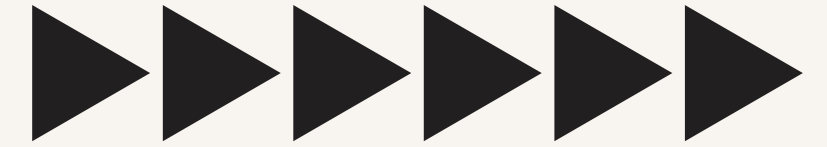
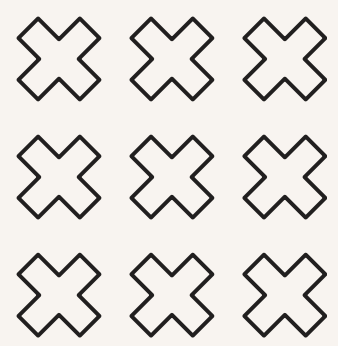




Real-Life Applications

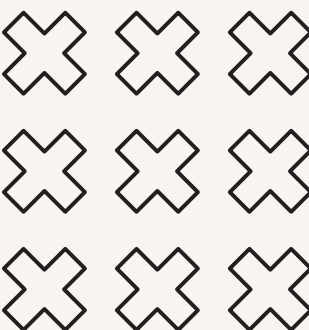
Women can utilize **emergency alerts** and **GPS tracking** in various scenarios, such as during travel, nightlife, or outdoor activities. Understanding how to implement these tools in daily life can significantly boost confidence and safety.

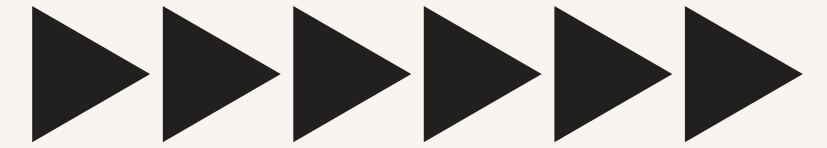
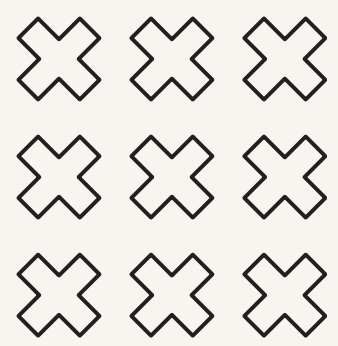




Challenges and Limitations

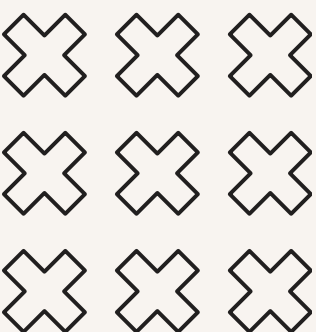
Despite the benefits, there are challenges with **emergency alerts** and **GPS tracking**, including **privacy concerns** and **technological barriers**. Addressing these issues is essential to ensure that women can use these tools effectively and without fear.





Empowering Women through Education

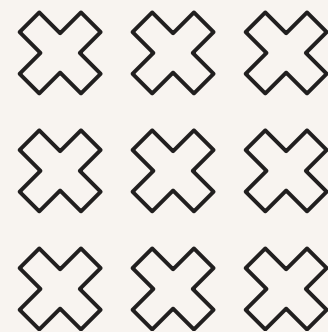
Education on the use of **emergency alerts** and **GPS tracking** is vital. Workshops and community programs can equip women with the knowledge to utilize these tools confidently, ultimately leading to greater personal safety and empowerment.





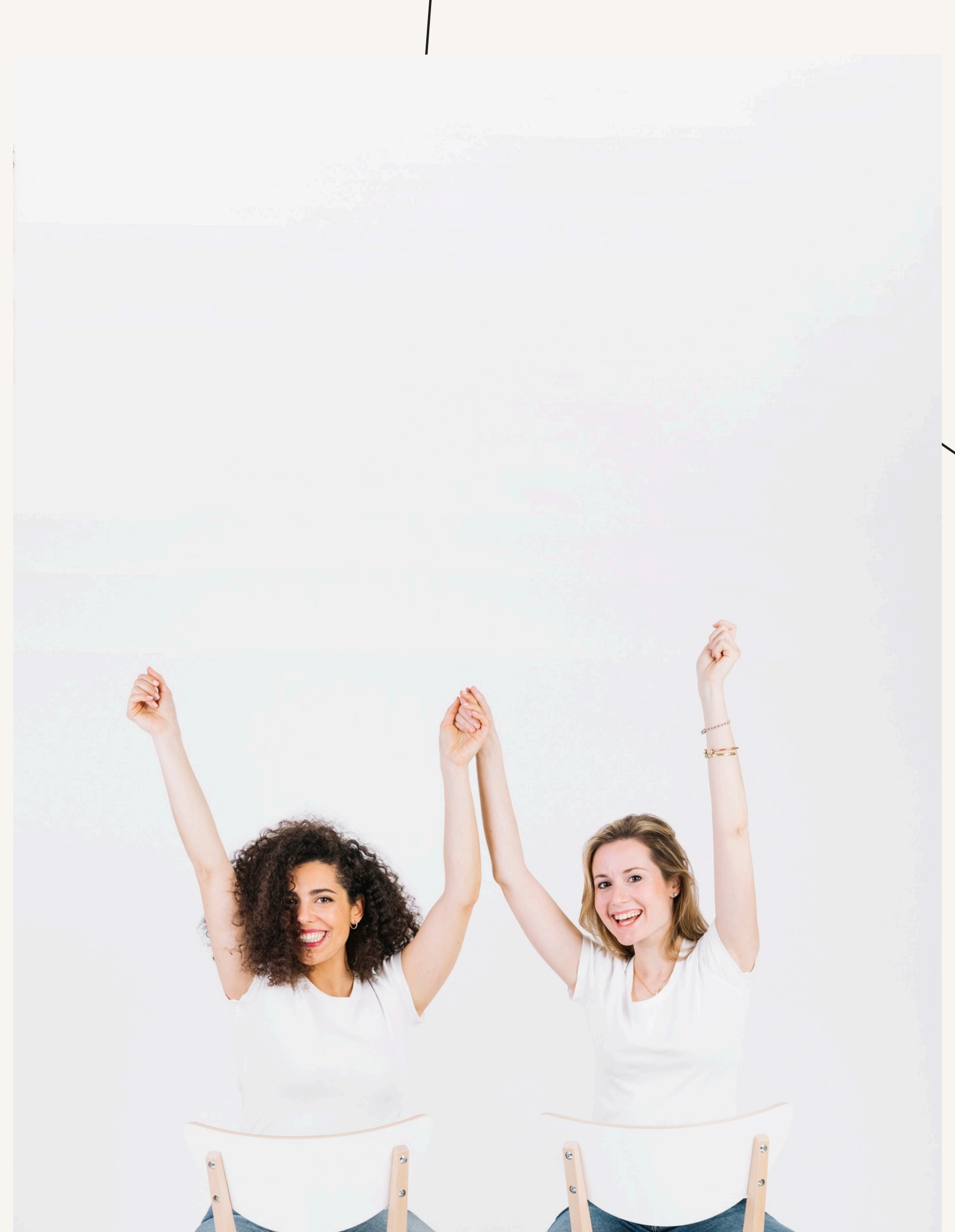
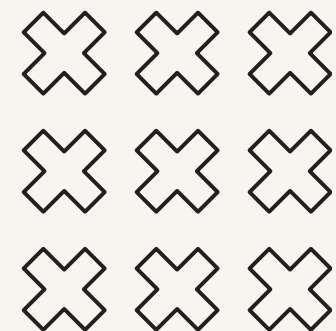
Future of Personal Safety Technology

The future of **personal safety technology** looks promising with advancements in **AI** and **wearable devices**. These innovations will likely enhance the effectiveness of **emergency alerts** and **GPS tracking**, providing even better solutions for women's safety.



Conclusion:

Empowering women through **emergency alerts** and **GPS tracking** is essential for enhancing personal safety. By leveraging technology and education, we can create a safer environment, allowing women to thrive and pursue their ambitions without fear.





Thanks!

